Blackout

System Operator · Calder Interconnection

15 DAYS · 52 STOPS · GRADE 12 · BLACKOUT

THE OPENING CARD

Calder Valley is heading into its coldest stretch of winter, and almost all of its outside power crosses one transmission corridor that already has little spare margin. You are the night system operator for a network serving four million people. A bad call can cut power to homes, a dialysis clinic and the valley pumps; a worse one can let a local failure spread. Across fifteen shifts you must keep supply and demand balanced, determine how hard the Calder corridor can safely be pushed, and write the restart plan before the winter case is signed. The catch is that every one of those decisions depends on numbers you may not be able to trust.

DAY 1 OF 15

Four degrees colder

BEFORE THE DAY — GREET

The shift changes at seven and nobody has met you

Two crews work this station and they hand over at seven. Farrow, the metering and standards engineer, wants you known to both before anything goes wrong, because at four in the morning a voice on a radio is either somebody you have met or somebody you are going to argue with. Get round the yard and put a name to most of the shift before the handover is done.

PLAN CARD 117W

Monday night. Tomorrow will be four degrees colder than anyone planned for, and Calder Valley already depends on a transmission corridor with little spare margin. Sten Lindgren can estimate the centre of the evening peak, but not exactly how high it will go. Rafael Alvarez has worse news: the battery being counted as reserve returned less energy than its brochure promised. Dolores Reyes has to sign tonight for how much power is held back instead of sold. Count reserve that is not really there and the missing power appears at the coldest hour, when the corridor is hardest to relieve and customers may have to be switched off. Today you decide what the system can honestly promise.

BRIEFING 19W

Tomorrow's cold peak will push the Calder corridor harder, and the battery returned less energy than the plan assumed.

WHAT YOU DECIDE 22W

Decide how much reserve is real enough to count when tomorrow is colder than planned and the battery delivered less than promised.

1.1 What comes out against what went in

LOAD & FORECASTING · A ROOM · BALLPARK

SCENE 46W

The battery is supposed to cover part of tomorrow's cold peak. Alvarez has the measured energy that went in Saturday night, the energy that came back Sunday, and the brochure rating. Reyes wants to know how much of tomorrow's reserve she can safely credit to it.

GUIDE 67W

Four numbers, and two of them are the power rating and the rated duration. Those say how fast and for how long, not how much came back. Ask of each whether it is an energy or a rate. A round trip is what went in against what came out, over a whole cycle. Three numbers on the desk disagree, and the brochure figure is one of them.

ASK 7W

Estimate the round-trip efficiency of the store.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

storing energy and recovering it both lose some of it · electrical power and energy over time — taken as read

VERDICT 76W

Everything that moves energy in and out loses a share of it, in the converter, in the chemistry and in keeping itself cool. Round-trip efficiency is the plain ratio of what came out to what went in, over a whole cycle. It is the only figure that tells you what the store is worth as reserve. A power rating is a different claim: it says how fast, not how much. Duration is a third one again.

TAKEAWAY 12W

Round-trip efficiency is measured on the energy, not on the power rating.

1.2 The reason the wires are not at wall voltage

GENERATION · GENERATION HALL · A PERSON · CHOICE

SCENE 50W

The replacement generation for tomorrow's peak is far from Calder Valley. Haddad points to the station transformer: the generator makes power at 20 kV, while the transmission line runs at 400 kV. Reyes wants to know what that voltage change buys before she counts the distant plant as useful reserve.

GUIDE 64W

All four options pair a change in current with a change in loss. Two of them get the current right and disagree about the heat. Ask what the loss depends on, and what power it depends on it to. Current falls in proportion to voltage. Heat follows the square of current. Work both halves before choosing, because three of these get one half right.

ASK 17W

For the same real power, what happens when voltage is stepped from 20 kV to 400 kV?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a transformer changes voltage in proportion to its turns · electrical power and energy over time — taken as read · faraday's law and electromagnetic induction — taken as read

VERDICT 78W

For a given real power, raising voltage allows the same power to move with less current. Resistive loss then falls with current squared. Halving current cuts that loss to one quarter. A transformer makes the trade possible by changing voltage in proportion to turns while transferring nearly the same power, apart from losses.

That is why long-distance transmission uses high voltage despite the harder insulation problem. The cost of insulation buys a much larger saving in conductor heating.

TAKEAWAY 18W

The same power can be delivered in more than one way, and the ways are not equally wasteful.

1.3 What a node has to add up to

SYSTEM OPERATIONS · A ROOM · CHOICE

SCENE 31W

The busbar meter reads 900 A out. Reyes wants to know what that single figure rules out about the two circuits underneath it. A printed network diagram lies on the desk.

GUIDE 63W

All four options say something about two parallel circuits. They differ in whether they need one equation or two. Ask of each whether the meter reading alone could establish it. Charge does not pile up at a junction, so the branch currents have to add to the total. That fixes their sum and nothing else. Two of these options confuse parallel with series.

ASK 12W

What does the 900 A total tell you about the 2 circuits?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

charge does not accumulate at a junction · ohm's law and resistive networks — taken as read · charge, current, voltage and resistance — taken as read

VERDICT 85W

Charge does not build up at a junction, so current entering a node must equal current leaving it. Kirchhoff's current law gives that conservation rule. It fixes the sum of the branch currents without telling you how the total divides. The split needs another relation, such as the branch impedances. Assuming an even split is wrong when the paths differ. Assuming each branch carries the full total confuses parallel and series circuits. One equation constrains two unknown branch currents; it does not determine both alone.

> **TAKEAWAY** 14W
>
> A conservation law bounds the parts of a system without identifying any of them.

END OF THE DAY 14W

Reserve is useful only if the energy is actually there when uncertainty becomes demand.

DAY 2 OF 15

Three things at once

PLAN CARD 102W

Tuesday, 16:40. Three problems land at once. A cable fault has cut power to 3,300 homes. Actual demand is already 200 MW above the forecast and the gap is still growing. The gas station carrying the evening peak must shut at 18:00 because it is running out of fuel. Each problem is manageable alone; together they can consume the reserve and push more power through the Calder corridor just as the cold evening arrives. Reyes can keep only one problem in her own hands. Today you decide which problem is stealing options fastest, and exactly how the other two are handed off.

BRIEFING 17W

A feeder fault, a 200 MW forecast miss and a fuel-limited plant collide before the evening peak.

WHAT YOU DECIDE 11W

Rank three simultaneous problems, and hold the reasoning while they interfere.

2.1 Which of the three cannot wait

SYSTEM OPERATIONS · A ROOM · DELEGATE

SCENE 36W

Reyes has three live problems on the board and only one attention channel she can keep herself. Aaron Whitlock is ready to take a handoff, but she refuses any instruction that amounts to simply “watch it.”

GUIDE 60W

Three live problems and one attention channel you can keep. For each of the other two, name an owner, the first real action they will take, and the reading or event that brings the problem back to you. Reyes will not accept anything that amounts to 'watch it'. Then take the watch on the one that needs your own judgement.

ASK 11W

What do you keep, and what exactly do you hand off?

BEHIND THE BUTTON 138W

Why 'watch it' fails. An owner who has been told to observe will come back at the moment they need instructions, which is the moment you are busiest. A first action is what lets somebody hold a problem without you, and naming it is the difference between delegating and deferring.

What a return condition is for. It is not a promise to check in. It is the threshold at which the problem stops being theirs, agreed in advance, so nobody has to judge in the moment whether this is bad enough to interrupt you.

How to choose what to keep. Compare the three on how fast they are getting worse, on whether they are already contained, and on what happens if nobody acts. The one that needs judgement rather than procedure is the one you cannot hand over.

TAKES AS READ

a stable problem can be severe without becoming more urgent each minute · reserve actions need lead time before the peak

VERDICT 86W

The cable outage is loud but stable: customers are already off and a crew is moving. The fuel-limited machine has a known time and quantity, so another operator can prepare its replacement. The forecast error is different. It is still growing, and reserve procurement needs lead time before the evening peak. Every update that passes without action removes options that cannot be recovered later. Delegation is therefore not “watch it.” Each handoff needs an owner, a first action and a condition that brings the problem back.

TAKEAWAY 22W

Urgency comes from how fast a condition is worsening, how reversible it is, and whether somebody else can own the next move.

2.2 Loss when the load turns up anyway

TRANSMISSION & PROTECTION · A PERSON · BALLPARK

SCENE 39W

With the feeder out, flow has redistributed and the corridor is carrying more. Novak, the protection engineer, wants the heat figure before he decides whether the evening needs a switching change. A printed network diagram lies on the desk.

GUIDE 66W

Five numbers, and two of them are the current an hour ago and the conductor temperature. Neither belongs in the loss. Ask of each number whether the heat depends on it. And note the square. Current has risen about eleven per cent, and the heat rises by nearly a quarter. The rating is thermal, so that is a bigger bite of the margin than it looks.

ASK 10W

Estimate the power now lost as heat in the corridor.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

loss in a conductor rises with the square of the current · ohm's law and resistive networks — taken as read · electrical power and energy over time — taken as read

VERDICT 72W

The same loss relation now gives a worse answer. Resistive heating rises with the square of RMS current, across all three phases. Current has risen by about 11%, from 1,150 A to 1,280 A. The heat rises by about 24%, from roughly 17 MW to 21 MW. That matters because the conductor rating is thermal. A small-looking increase in current can therefore use a much larger fraction of the available temperature margin.

TAKEAWAY 17W

When a quantity enters squared, a small change in it is a large change in the result.

2.3 An order that survives being carried out

LOAD & FORECASTING · A ROOM · SEQUENCE

SCENE 32W

Lindgren has four moves for the evening and no view on the order. Two of them change what the others are worth, and one of them cannot be reversed inside the shift.

GUIDE 62W

All four moves are available right now, so this is not a clock. Ask of each two things. What does it improve about the numbers you are working from? And how easily can it be taken back? One of these sizes every other figure in the shift. One of them costs money rather than customers, and one takes real minutes to load.

ASK 10W

Order the evening's moves so none of them undoes another

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

some actions change the value of the actions that follow them

VERDICT 73W

Correct the forecast first. Every other number is sized against it, and a wrong demand figure means doing the right thing to the wrong amount. Bring reserve forward next. It is reversible, it costs money rather than customers, and it covers the machine coming off at 1800. Then move the replacement plant, which takes real minutes to load and has to be underway before the fuel limit bites. Reconfiguring the network is last.

TAKEAWAY 15W

Do the moves that improve your information first and the moves you cannot undo last.

END OF THE DAY 22W

Ranking is not about which problem is worst but about which one gets worse fastest and which one is cheapest to remove.

DAY 3 OF 15

The number on the wall

PLAN CARD 103W

Wednesday, 04:12. The biggest power station on the system trips without warning. Frequency falls, then settles low. Reyes says the flattening is not a recovery: it means the system has found a new balance at the wrong frequency. Haddad can start more generation, but the first clean clue to how much vanished is already disappearing from the frequency trace. Whatever replaces the missing power will also change the flow through the Calder corridor. Today you use the first seconds of the event to measure the shortfall before slower controls blur the evidence, and you decide which records are trustworthy enough to act on.

BRIEFING 15W

The biggest station tripped at 04:12; frequency has stopped falling, but it has not recovered.

WHAT YOU DECIDE 21W

Use the first seconds after a generator trip to determine how much power vanished and what the low frequency actually means.

3.1 What a frequency trend is evidence of

SYSTEM OPERATIONS · A ROOM · PROTOCOL

SCENE 36W

The wall shows frequency dropping for eight seconds, then flattening low. Aaron Whitlock, the assistant operator, has the log open. Two people have already named a cause, and he wants a statement the record can defend.

GUIDE 62W

Four observations, and the right column says what each one is evidence of. Pair them by asking one question. Is this quantity system-wide, or local to a place? Frequency is one number for the whole interconnection. Voltage is not. That difference is what lets a trend say whether something happened, while a voltage says where. Two people have already named a cause.

ASK 10W

From a falling trend to a statement about the system

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

an alternating supply has a frequency measured in cycles per second · synchronous machines and synchronisation — taken as read · electrical power and energy over time — taken as read

VERDICT 80W

System frequency reflects the balance between real power entering the interconnection and real power being demanded. After a sudden generation loss, stored kinetic energy in synchronized machines supplies the deficit briefly, so frequency begins to fall. The initial rate of change depends on both the power mismatch and that stored energy. A later flattening can be governor response at a lower frequency, not full recovery. Frequency is therefore strong evidence of a system-wide imbalance, while voltage is much more local.

TAKEAWAY 20W

Frequency is a system-wide balance signal; its rate of change depends on both the power mismatch and stored kinetic energy.

3.2 How much power actually left the system

GENERATION · GENERATION HALL · A PERSON · BALLPARK

SCENE 33W

Haddad, the generation lead, needs a number before starting anything. The only clean evidence is the first two seconds of the trend and the stored-energy figure for the machines that were online overnight.

GUIDE 66W

Five numbers, and two of them belong to the aftermath rather than the first instant. The settled deviation is what the governors did afterwards. The generation online is the denominator of a different question. Ask of each number whether it describes the first two seconds. A figure that is right about a later moment is wrong here, and the control room acts on whatever comes out.

ASK 10W

Estimate the generation lost at the moment of the trip.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a rate of change can be read off the steep part of a trend · synchronous machines and synchronisation — taken as read · electrical power and energy over time — taken as read

VERDICT 71W

Immediately after a generator trip, governor and reserve actions have not yet changed the power balance. Stored kinetic energy supplies the shortfall for the first moments, so frequency moves. With stored kinetic energy E_k, nominal frequency f_0 and measured rate df/dt, the imbalance is $\Delta P = 2E\_k|df/dt|/f_0$. Using 25,000 MW·s, 50 Hz and 0.30 Hz/s gives about 300 MW. This estimate is most useful before slower controls begin reshaping the trace.

TAKEAWAY 15W

The initial rate of frequency change measures the size of the imbalance, given the inertia.

3.3 The order the records claim things happened in

METERING & STANDARDS · A ROOM · TRACE

SCENE 31W

June Farrow, the metering engineer, has three records of the same eight seconds. They disagree about what happened first. One record was stamped by a clock nobody has checked since spring.

GUIDE 56W

Open each record to see which clock and which physical input it depends on. Keep the ones that would still stand if one clock were wrong, and untick the rest. Then name the shared source behind the false early timestamps. Three records disagreeing about the order is evidence about the records, not only about the event.

ASK 16W

Which records still support the event order, and what shared source explains the false early timestamps?

BEHIND THE BUTTON 141W

Why a clock is part of the measurement. A record's timestamp is as good as the clock that made it, and a clock nobody has checked since spring can be seconds out. Every event stamped by it moves together, which is exactly what makes a false ordering look consistent.

How to tell dependence from agreement. Two records agreeing prove nothing if both were stamped by the same clock. What matters is whether a record's timing came from somewhere else — a synchronised source, a physical input with its own time base — because only those constrain the order independently.

What the order is for. Sequence-of-events analysis is how a blackout is explained: what tripped first decides whether this was a protection failure, an operator action or a cascade. Getting the order wrong produces a confident and wrong account of the night.

TAKES AS READ

a timestamp combines a measured event with a clock reference · two channels can agree because they share the same source · faraday's law and electromagnetic induction — taken as read · rMS against peak and average — taken as read

VERDICT 80W

A timestamp contains a physical event and a clock claim. Those can fail separately. Two relay-derived records can agree because they share one internal clock, so their agreement is not two independent measurements. The GPS disturbance recorder uses a different time source and preserves the physical ordering. The slower SCADA record can still confirm breaker state, even though its two-second scan cannot resolve sub-second order. Dependency tracing keeps useful evidence instead of discarding every channel touched by the bad clock.

TAKEAWAY 14W

Agreement is independent evidence only when the channels do not share the same dependency.

3.4 What comes out against what went in — Review

LOAD & FORECASTING · A ROOM · CALLBACK · BALLPARK

SCENE 39W

Sten Lindgren, the load forecasting analyst, has a second store's Thursday cycle open: one charge overnight, one discharge into the morning rise. Rafael Alvarez, the demand analyst, wants to know whether it is worth counting as reserve at all.

GUIDE 84W

Four numbers, and only two of them are energies. The other two say how hard the unit can push and for how long it can keep pushing, which is a different claim about the same machine. Ask of each tile what it would be measured in. A round trip is one energy divided by another energy of the same kind, over one complete cycle. This unit is smaller and older than the one on Sunday's log, and nobody has cycled it deliberately since spring.

ASK 8W

Estimate the round-trip efficiency of the older store.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

storing energy and recovering it both lose some of it · a rating in megawatts is a rate rather than an amount · electrical power and energy over time — taken as read

VERDICT 80W

Every store loses a share of what it is given, in the converter, in the chemistry and in holding itself at temperature, and an older unit loses more. The test does not change with the machine. Divide what came back out by what went in, over one complete cycle. The answer is the fraction of the nameplate you may count as reserve. A rating in megawatts answers how fast, and a duration answers how long. Neither answers how much survives.

TAKEAWAY 21W

A store's worth as reserve is a fraction, and the same fraction is asked of every machine whatever its nameplate says.

END OF THE DAY 20W

Frequency is not a reading from a place; it is the whole system's supply and demand, arriving as a number.

Forty minutes

BEFORE THE DAY — FOLLOW

The switching route, walked once by somebody who knows it

Obi, the distribution lead, walks the restoration route the way it has to be walked, in the order the isolations happen, and she will not do it twice. Stay with her. Too far back and you lose which bay she turned into; too close and you are standing in front of the cubicle she is reading. She stops at each isolation point long enough to look at it, and that is where the walk is actually learned.

PLAN CARD 101W

Thursday. Replacement power from Wednesday's trip is being pushed through the two Calder circuits, and both are running hot. Novak says the operating model gives the corridor a little emergency margin before protection takes it out. Farrow points out that the margin comes from a temperature record nobody has independently checked. Dube is in the yard waiting for a switching order. Today you decide how early to move generation away from Calder and how hard the corridor can be pushed while that move arrives. If protection opens the corridor first, the valley loses the choice of how the next minutes happen.

BRIEFING 20W

The Calder corridor is above its continuous rating, and the room believes it has about forty minutes of emergency margin.

WHAT YOU DECIDE 17W

Decide whether the Calder corridor can be held safely, and act before its emergency margin is gone.

4.1 What you do before a relay does it for you

TRANSMISSION & PROTECTION · A ROOM · TRIGGER

SCENE 34W

The corridor is at 108% of continuous rating and rising slowly. Novak, the transmission engineer, wants a response he can hand to Dube, the switching operator, before the relay makes the choice for them.

GUIDE 69W

The live trend is rising about 1.5 percentage points of continuous rating per hour. Redispatching 200 MW away from Calder takes six hours from the call, and 120% is the emergency ceiling. Work backward from 120 by the rise expected during those six hours. The rule has to be written before the trend is released, because a threshold chosen after seeing the peak is hindsight, not an operating rule.

ASK 20W

At what loading do you commit the redispatch, given the move takes six hours and protection acts at 120 %?

BEHIND THE BUTTON 140W

Why act before the relay. Protection will disconnect the corridor when its limit is reached, and it will do so without regard to what else that leaves connected. Anything you do before then is a controlled choice; anything after is a consequence.

Why two thresholds rather than one. Redispatching generation takes minutes and shedding load takes seconds. A single threshold set for the slower action fires far too early for the faster one, and set for the faster one leaves no time for the slower. The lead time is the reason there are two numbers.

What Dube needs from this. A switching operator acting at three in the morning needs a rule, not a judgement. Numbers agreed in advance are what make the night's actions defensible afterwards, and what stop the decision being made by whoever happens to be awake.

TAKES AS READ

continuous and emergency ratings are different limits · redispatch is slower than automatic or operator load shedding · charge, current, voltage and resistance — taken as read · ohm's law and resistive networks — taken as read

VERDICT 63W

The emergency ceiling is not the point at which a slow action should begin. At roughly 1.5 percentage points per hour, six hours consumes about nine percentage points of loading margin. Working backward from 120% therefore puts the redispatch trigger near 111%. The calculation matters because an action can be physically correct and still be useless if it starts too late to arrive.

TAKEAWAY 20W

A protection plan must be written before the number moves, with earlier thresholds for actions that take longer to arrive.

4.2 The power that never arrives

LOAD & FORECASTING · A ROOM · BALLPARK

SCENE 32W

Sten Lindgren, who forecasts demand, wants the loss figure before the morning call. The argument about moving generation is really an argument about waste. The yard radio is still live beside them.

GUIDE 61W

Five numbers, and two of them are the voltage and the power factor. Neither appears in a resistive loss. Ask of each number whether the heat depends on it. Heat comes from current and resistance, and it is paid once per phase. Note the square while you are choosing. Ten per cent more current is about twenty-one per cent more heat.

ASK 10W

Estimate the power lost as heat in the loaded corridor.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

current through a resistance dissipates power as heat · charge, current, voltage and resistance — taken as read

VERDICT 81W

Resistive loss in one phase is the RMS current squared times that phase resistance. A balanced three-phase corridor has three such conductors, so the total is three times I squared R. The square is the key. 10% more current means about 21% more heat. High transmission voltage helps because the same real power can be moved with less current. Lower current then cuts resistive loss sharply, which is why a modest overload can consume thermal margin faster than its percentage suggests.

TAKEAWAY 18W

Loss rises with the square of current, which is why a small overload costs more than it looks.

4.3 What the path opposes at 50 hertz

SYSTEM OPERATIONS · A PERSON · BALLPARK

SCENE 43W

Reyes has the switching order for the second circuit in front of her. Novak has read the path's own figures out of the records: 3.1 ohms of resistance, and 12.4 ohms of reactance at 50 hertz. She wants one number before she signs.

GUIDE 85W

Five numbers, and only two of them are properties of the path. Ask of each whether it describes what the path is, or what is being pushed through it. Resistance and reactance are both measured in ohms and both oppose the current, but they do it at different moments in the cycle, so they do not add end to end. Combine them the way you would two sides of a right-angled triangle, and notice before you commit how little the smaller one moves the answer.

ASK 9W

Estimate what the corridor path opposes at 50 hertz.

BEHIND THE BUTTON 190W

Why the two do not simply add. Resistance turns electrical energy into heat and opposes the current itself. An inductance stores energy in a magnetic field and hands it back a quarter of a cycle later, so what it opposes is the change of current. The two oppositions peak at different instants, and adding them arithmetically would count a maximum that never happens.

Why an overhead path is mostly reactance. A long circuit is a large loop of wire, and its inductance grows with the area that loop encloses. At the spacings a 400 kV tower needs, that inductance dominates: on this path the reactance is four times the resistance, so the resistance contributes about three per cent of the total, and heat is a small part of the story.

What the frequency has to do with it. Reactance is proportional to frequency, so a figure like this one is only true at the frequency it was measured at. Resistance is nearly a property of the metal; reactance is a property of the metal and the cycle together, which is why a record of it has to say 50 hertz.

TAKES AS READ

an alternating current has a frequency, and this system runs at 50 hertz · the hypotenuse of a right-angled triangle is the square root of the sum of the squares · aC waveforms: frequency, period and phase — taken as read · capacitance, inductance and stored energy — taken as read · rMS against peak and average — taken as read

VERDICT 84W

Resistance opposes the current itself and turns part of it into heat. Reactance opposes the change of current, because an inductance answers a rising current by pushing back, and it does that a quarter of a cycle away from the resistance. Two oppositions that peak at different moments cannot be added end to end; they combine as the two shorter sides of a right-angled triangle. Here the reactance is four times the resistance, so the resistance moves the total by about three per cent.

TAKEAWAY 22W

Two oppositions that peak a quarter of a cycle apart combine at right angles, so the larger of them dominates the total.

4.4 The reason the wires are not at wall voltage — Review

GENERATION · GENERATION HALL · A ROOM · CALLBACK · CHOICE

SCENE 42W

The spare bank at the far end of the yard has been left energised through three quiet nights so it can be picked up instantly. Nadia Haddad, the generation lead, wants the standing cost of that decision written down before the audit.

GUIDE 73W

All four options divide the loss into two parts and disagree about which part the load controls. The test is what each part is dissipated in, and what has to be present for it to appear at all. Work out what is still going on inside a transformer that is connected and carrying nothing. The distinction decides whether leaving spare plant hot is free or is a bill every night of the year.

ASK 15W

A transformer bank sits energised all night carrying almost no load. What is it costing?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a transformer is energised the moment it is connected, whether or not it carries load · electrical power and energy over time — taken as read · faraday's law and electromagnetic induction — taken as read

VERDICT 79W

A transformer has two families of loss and they answer to different things. Copper loss is dissipated in the windings by the current through them, so it rises and falls with the load and disappears when the load does. Iron loss is dissipated in the core by the flux reversing fifty times a second, and the flux is set by the applied voltage rather than by the load. Energise the bank and you pay it, all night, carrying nothing.

> **TAKEAWAY** 23W
>
> A cost that does not follow the load is still a cost, and it is the one an idle machine goes on paying.

END OF THE DAY 26W

A thermal rating is a temperature under stated conditions, not a magic current wall; an action threshold must leave enough time for the action to arrive.

Who gets power first

PLAN CARD 106W

Friday. The big lines are steady again. Eleven street-level circuits are still dead. Chinelo Obi, the distribution lead, has three crews and fourteen thousand homes with no power. One circuit feeds a dialysis clinic. Another feeds a thousand houses. Ewa Kowalczyk, the field crew lead, is at the first fault, and it is a fallen tree. The far end of one long circuit is running below its proper voltage, and nobody in the room can say why. Today you pick which circuit gets the first crew. Every hour is somebody sitting in a cold house, and an inquiry will read your order back to you later.

BRIEFING 19W

Eleven feeders are still dead; three crews cannot reach all of them, and one feeder serves a dialysis clinic.

WHAT YOU DECIDE 15W

Restore customers in an order that protects the people for whom waiting costs the most.

5.1 What is common and what divides

DISTRIBUTION · DISTRIBUTION DEPOT · A ROOM · PROTOCOL

SCENE 30W

Obi, the distribution supervisor, is training a new starter. Four circuit sketches are on the whiteboard, and he wants each matched to the quantity that stays common across that arrangement.

GUIDE 61W

Four arrangements, and the right column names what stays common in each. Pair them by asking how many paths the current has. One path means one current and a divided voltage. Two nodes shared means one voltage and a divided current. Then ask what adding a branch does to the total. Getting this wrong sizes a fuse for the wrong current.

ASK 7W

Match each connection to what it shares

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a circuit can be connected end to end or side by side · kirchhoff's laws and nodal analysis — taken as read

VERDICT 86W

Series and parallel circuits differ by what stays common. In series there is one path, so every element carries the same current. The supply voltage divides across the elements. In parallel every branch connects to the same two nodes, so each branch has the same voltage. The current divides instead. Adding a parallel branch lowers the equivalent resistance and raises total current. Adding a series resistance raises the equivalent resistance and lowers the loop current. Mixing these rules leads directly to wrong fuse and conductor sizing.

TAKEAWAY 16W

Series shares the current and divides the voltage; parallel shares the voltage and divides the current.

5.2 What actually limits a conductor

TRANSMISSION & PROTECTION · A PERSON · PROTOCOL

SCENE 35W

Novak is asked to allow more current down a circuit for the afternoon. He wants each reason for saying no matched to the thing it is actually about. The shift log is open beside them.

GUIDE 63W

Four observations, and each is limited by a different thing. Ask of each whether it is set by current, by voltage, or by geometry. Heat comes from current and does not care about voltage. Insulation cares about voltage and not about current. A volt drop is a customer problem. One of these four is where a heat limit turns into a safety one.

ASK 7W

Match each observation to what it constrains

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

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TAKES AS READ

a conductor warms up as current passes through it · electrical power and energy over time — taken as read

VERDICT 83W

Two different limits get confused. Heat comes from current, and it does not care what voltage the conductor is at. The insulation and the clearances care about voltage, and they do not care what current is flowing. So a circuit can be at nominal voltage and still be over its thermal limit, which is the case in front of the room. Volt drop is a third thing again, a consequence of current, and it is a customer problem rather than a safety one.

TAKEAWAY 14W

> Current sets the heat, voltage sets the insulation, and the two limits are separate.

5.3 The order somebody will read back to you

SYSTEM OPERATIONS · A ROOM · CHOICE

SCENE 40W

Three crews, eleven dead feeders, and a list in front of Reyes that mixes household counts with one dialysis clinic and one pumping station. The control-room phone is already ringing. The shift log and network board are open beside them.

GUIDE 61W

Four feeders, and household count is only one of two things that matter. Ask of each what waiting costs, and how long the work takes. A one-hour job frees the crew for the next fault this shift. A four-hour job does not. One of these is measured in treatments rather than in inconvenience, and one is already on a standby supply.

ASK 7W

Which feeder should take the first crew?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a crew can only work one fault at a time

VERDICT 75W

Two things decide this, and household count is only one of them. The first is what waiting costs, and for a dialysis clinic it is measured in treatments rather than in inconvenience. The second is how long the work takes, because a one-hour job frees the crew to take the next fault the same shift. The clinic circuit is worst on the first and best on the second, so it goes first on both counts.

TAKEAWAY 17W

The first crew goes where the consequence of waiting is worst, not where the count is biggest.

END OF THE DAY 16W

Restoration is a series of choices about people, made with instruments that only see the substation.

Can we trust the numbers?

PLAN CARD 93W

Monday. Wednesday night has to be written up, and Reyes wants it filed today. Farrow will not sign it. Two of its numbers come from instruments last checked in the spring. Nadia Haddad needs a machine back on the system before the evening peak, and there is a correct order for switching it in. Today you work out what the system was really delivering, and settle a fight between two instruments that disagree. Farrow is right to wait. A number in a report lasts longer than the people who knew what it meant.

BRIEFING 20W

The report is due, but two key readings have not been independently checked and two calibrated instruments appear to disagree.

WHAT YOU DECIDE 17W

Decide which measurements the incident report can rely on, and which apparent disagreements are only different definitions.

6.1 What three phases actually add up to

METERING & STANDARDS · A ROOM · BALLPARK

SCENE 36W

Farrow has line voltage, line current and a power factor from the night of the event. The report claims a figure for delivered power that nobody has reproduced. A printed network diagram lies on the desk.

GUIDE 71W

Six numbers, and two of them are traps that look right. There are three phases, and the factor is not three. And the sine of the angle is not what selects real power. Ask of each number what definition it belongs to. A voltage measured between two phases differs from one measured to neutral, and that is where the factor comes from. A report already claims a figure nobody has reproduced.

ASK 8W

Estimate the real power the circuit was delivering.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a three-phase supply carries power on three conductors at once · electrical power and energy over time — taken as read

VERDICT 72W

The tempting mistake is to multiply by three because there are three phases. The factor is root three instead. Line voltage is measured between two phases, not from one phase to neutral. Those voltage definitions differ by root three, so the line-quantity power formula carries that factor. Power factor then selects the in-phase current that produces real power. Using three instead of root three overstates the answer enough to change the report.

TAKEAWAY 16W

Three-phase power from line quantities carries a factor of root three, and it is not three.

6.2 Four things that have to agree before the breaker closes

GENERATION · GENERATION HALL · A PERSON · SEQUENCE

SCENE 32W

Haddad has a machine at rest and a system running at nominal. She wants the sequence written down before anybody is standing at the panel. The next operating call is minutes away.

GUIDE 61W

All four of these happen at the panel, so this is not a matter of what is available. Ask of each what mismatch it removes, and what that mismatch would do if the breaker closed with it still there. A speed difference and a voltage difference cause different damage. One of these four is the consequence of the others being right.

ASK 10W

Put the steps of synchronising in the order physics requires

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a generator joined to a live system is forced to turn at the system's speed · aC waveforms: frequency, period and phase — taken as read · faraday's law and electromagnetic induction — taken as read

VERDICT 80W

Each step removes a different mismatch before the breaker closes. Match frequency first, because a speed difference forces a violent torque when the machine is tied to the grid. Match voltage next to avoid a large reactive current. Then wait for phase angle to approach zero as the two waveforms drift past each other. The breaker closes only when frequency, voltage and phase are all inside their allowed windows. Closing earlier turns a synchronization error into electrical and mechanical stress.

TAKEAWAY 11W

Speed, then voltage, then phase angle, and only then the breaker.

6.3 Two instruments, two quantities

TRANSMISSION & PROTECTION · A ROOM · CHOICE

SCENE 35W

An oscilloscope on the busbar shows a peak of 325 volts. The panel meter beside it reads 230. Both were calibrated last month and neither has moved. The yard radio is still live beside them.

GUIDE 63W

Four relations, and both instruments are in calibration. So the question is what each number is a summary of. An alternating voltage has no single value. One instrument reports the highest the waveform reaches. The other reports the steady voltage that would heat a resistor the same. Ask which way the conversion runs before choosing, because two of these options have it backwards.

ASK 10W

Which numerical relation should hold if both instruments are correct?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

an alternating voltage changes continuously through each cycle · aC waveforms: frequency, period and phase — taken as read

VERDICT 88W

An alternating voltage does not have one value, so any single number is a summary and the summary has to be named. The peak is the highest the waveform reaches. The RMS value is the steady voltage that would deliver the same power into a resistance. For a sine wave it is the peak divided by root two. Divide 325 by 1.414 and the meter's 230 comes straight out. This is why mains voltages are quoted as RMS everywhere: it is the figure that predicts heating and power.

TAKEAWAY 15W

> Peak and RMS are different quantities, and one is the other divided by root two.

6.4 What a node has to add up to — Review

SYSTEM OPERATIONS · A ROOM · CALLBACK · PROTOCOL

SCENE 44W

A relay upgrade has left one branch meter dead on a four-way node, and the yard wants to know whether the shift can carry on without it. Aaron Whitlock, the assistant operator, has the four readings in front of him, one of them blank.

GUIDE 69W

Four observations from the same substation, and the right column says what each one establishes. Pair them by asking which of the two bookkeeping rules is being invoked, and whether the observation settles anything on its own. One rule fixes a sum at a junction. The other fixes a sum around a closed path. What neither of them gives you is how a total divides between two ways round.

ASK 8W

Say what each observation at the node establishes

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

charge does not accumulate at a junction · a network drawing can be read as a set of junctions and closed paths · charge, current, voltage and resistance — taken as read · ohm's law and resistive networks — taken as read

VERDICT 79W

Both of these rules are bookkeeping, and bookkeeping is what makes a missing entry recoverable. Currents at a junction have to sum to nothing, so three branch readings and the incoming total give the fourth without a meter on it. Voltages around a closed path have to sum to nothing, so a path that does not close is a statement about the readings rather than about the network. Neither rule says how a total divides. That takes the impedances.

> **TAKEAWAY** 19W
>
> A rule that fixes a sum is how a term nobody measured is recovered from the terms somebody did.

END OF THE DAY 17W

A reading is a claim about one quantity, and two instruments can disagree while both are right.

The easier path

PLAN CARD 85W

Tuesday, a week on. The Calder corridor has two parallel circuits, but that does not mean the current splits in half. The newer circuit has about half the impedance of the older one, so it is carrying roughly twice the current and using thermal margin much faster. Dube is waiting to hear which circuit can be taken out for maintenance. Today you calculate the split before anybody opens a breaker. Take out the wrong path and the circuit left behind inherits all of the flow.

BRIEFING 18W

Two circuits share the Calder corridor, but the lower-impedance one is carrying twice the current of the other.

WHAT YOU DECIDE 17W

Work out how the two Calder circuits actually share current before one of them is switched out.

7.1 How the flow splits between two paths

TRANSMISSION & PROTECTION · A ROOM · BALLPARK

SCENE 34W

900 amps leave the busbar for the same destination by 2 routes. The older circuit is six ohms, the newer one three, and only the total is metered. The control-room phone is already ringing.

GUIDE 60W

Four numbers, and one of them is the busbar voltage, which cancels out of a divider. Both circuits connect the same two busbars, so they see the same voltage. Ask of each number whether it describes this path, the other path, or both. Current divides against impedance, so the path being asked about is not the impedance the formula wants.

ASK 9W

Estimate the current flowing in the older, higher-impedance circuit.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

two paths between the same two points carry the same voltage across them

VERDICT 63W

Both circuits connect the same two busbars, so they have the same voltage across them. Current therefore divides inversely with impedance. With 6 Ω and 3 Ω, the lower-impedance circuit carries twice the current of the older one. The total busbar current fixes the sum; the impedances determine the share. This is why adding a parallel circuit does not guarantee an even split.

TAKEAWAY 10W

The lower-impedance path takes the larger share, in inverse proportion.

7.2 What is worth knowing before the peak

SYSTEM OPERATIONS · A PERSON · VALUE

SCENE 35W

Reyes has a small budget and four things she could find out before the evening peak. Only some of them would change what she does about the reserve. The shift log is open beside them.

GUIDE 57W

Five evidence credits, four things you could find out, and a reserve decision at six o'clock. Open each card and ask what its result would change. A measurement that confirms something already settled is not worth a credit, however clean it would be. Buy the ones that could still move the decision, and commit before the peak.

ASK 7W

Which measurements are worth buying before 18:00?

BEHIND THE BUTTON 145W

What makes evidence worth its credit. Two possible results that point at different actions. If both outcomes leave you doing the same thing about the reserve, the measurement is interesting and not useful — and tonight the difference between those two words is a credit you cannot get back.

Why the deadline is part of the question. A result that arrives at seven is a result about a decision already made. So a slower, better measurement can be worth less than a rough one in time, which is the opposite of how the same choice would be scored in a laboratory.

What the shift log is doing open beside her. It records what has already been decided and why, which is the fastest way to see which of the four questions are still open. Half of buying evidence well is knowing what is already settled.

TAKES AS READ

reserve is held against uncertainty rather than only the forecast centre · storage power and storage deliverability are separate claims · electrical power and energy over time — taken as read

VERDICT 86W

Reserve depends on uncertainty in demand and on resources that can actually deliver when called. Measuring this winter’s cold-load sensitivity can change the forecast width before the peak. Re-testing storage at the relevant discharge rate can change how much reserve the store deserves credit for. Repeating a generator test that is already current mainly buys reassurance. A feeder survey that arrives in six weeks may be useful planning evidence, but it cannot change tonight. Evidence has value only when its result can still alter the decision.

TAKEAWAY 18W

The value of a measurement is the decision it can change, not how precise or interesting it sounds.

7.3 Why the far end sits low

DISTRIBUTION · DISTRIBUTION DEPOT · A ROOM · BALLPARK

SCENE 34W

The substation end of the feeder is at nominal voltage. Customers four kilometres along it are complaining, and the only meter is back at the substation. The room is waiting for a written instruction.

GUIDE 61W

Six numbers, and one of them is the feeder's nominal voltage, which the drop does not use. Two pairs go together: resistance with the in-phase part, reactance with the out-of-phase part. Ask of each number which half of the impedance it belongs to. And note the rating. Eleven kilovolts is measured between lines, which is why a factor arrives with it.

ASK 11W

Estimate the volt drop from the substation to the far end.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

current flowing through a resistance produces a drop in voltage

VERDICT 81W

A real feeder has both resistance and reactance. Current through those terms produces a voltage drop whose size depends on the load phase angle. The resistive part follows the in-phase current. The reactive part follows the out-of-phase current. For a balanced three-phase feeder expressed with line-to-line voltage, the result carries a factor of root three. That factor matters here because 11 kV is a line-to-line rating. Reactive support can raise the far-end voltage by reducing the reactive part of the drop.

TAKEAWAY 15W

Volt drop is set by the current and by both parts of the line's impedance.

END OF THE DAY 14W

Current divides between parallel paths in inverse proportion to their impedance, whatever anybody intended.

80 milliseconds

BEFORE THE DAY — HUNT

Six earths, and the log says five

Before the line can be re-energised every temporary earth has to be off. The log says five were applied. Lindgren, the load forecasting analyst, counted six going on. Until all of them are found and accounted for nothing gets switched in, because an earth left on a line that is made live is a fault the size of the station. Find them.

PLAN CARD 111W

Wednesday. A cable failed inside the substation at 13:40. The protection cut it off in less than a tenth of a second. Nobody in the room saw the fault happen. They only saw what was left. Novak will not sign the switchgear back into use until somebody works out how big the current was. Farrow wants to know how an instrument reads a wire it never touches. Kowalczyk has a crew at the gate and an earthing rule they are impatient with. Today you work out the fault current, and decide who is allowed near it. The numbers here are huge and the times are tiny, so nobody guesses either one.

BRIEFING 17W

A cable failed at the substation and the protection cleared it before the alarm reached the room.

WHAT YOU DECIDE 11W

Establish what a fault does before anybody can respond to it.

8.1 What a short circuit is limited by

TRANSMISSION & PROTECTION · A ROOM · BALLPARK

SCENE 31W

The cable failed phase to earth. Phase voltage is 6,350 volts and the total impedance back to the source is 0.42 ohms. Novak wants the current the switchgear had to interrupt.

GUIDE 59W

Four numbers, and two of them belong to the healthy circuit and the protection. Normal load current is set by equipment drawing power, and a fault bypasses that equipment. Ask of each number whether it describes the load or the path back to the source. The clearing time decides how long the current flows, not how large it is.

ASK 8W

Estimate the current that flowed during the fault.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a short circuit leaves almost no impedance in the path

VERDICT 81W

Normal load current is set by the equipment drawing power. A fault bypasses that load, so the current is instead limited mainly by the source and network impedance back to the fault. Using the phase voltage and the Thevenin impedance gives the first symmetrical RMS estimate. Here 6,350 V divided by 0.42 Ω is about 15 kA. That is why switchgear has a separate interrupting rating. Clearing time then controls how long the large current can deposit heat and mechanical stress.

TAKEAWAY 17W

Fault current is the driving voltage over the impedance of the path, and nothing else limits it.

8.2 A reading taken without a connection

METERING & STANDARDS · A ROOM · CONTROL

SCENE 35W

June Farrow, the metering engineer, has a protected low-voltage current-transformer test loop beside the yard diagram. The bench shows primary current, secondary current and a movable core. She wants a cause, not a remembered definition.

GUIDE 66W

The rig is a current transformer: a conductor carrying current, and a core around it with a winding that reports a scaled copy. The number you are watching is that secondary current. Three controls can be changed, and a response counts only if it is larger than the bench's own wander. Change one, run it, put it back, and name the control the secondary current follows.

ASK 17W

Which change makes the secondary current disappear, and does the signal return when that change is reversed?

BEHIND THE BUTTON 173W

How a current transformer works, and why nothing touches the conductor. Current in the primary makes a changing magnetic field, the core carries that field around the winding, and the changing field induces a proportional current in it. Nothing is connected to the primary at all — which is exactly why the instrument is safe on a live circuit, and why what it needs is change rather than contact.

Why a steady current reads nothing. Induction responds to a field that is changing. Hold the primary current constant and the field stops changing, so the winding reports nothing while the conductor is still carrying full load. An instrument that reads zero on a live conductor looks broken and is behaving exactly as designed.

Why the core is the third control. Open the magnetic path and the field no longer links the winding, so the secondary falls away even with a changing primary. Two different controls therefore kill the reading for two different reasons, and telling them apart is what the reversal is for.

TAKES AS READ

a changing magnetic field can induce voltage in a nearby coil · the bench current transformer has 1 primary turn and 20 secondary turns · aC waveforms: frequency, period and phase — taken as read · charge, current, voltage and resistance — taken as read

VERDICT 82W

A current transformer needs no electrical connection between primary and secondary. Alternating primary current creates changing magnetic flux in the core. Faraday’s law then induces an emf in the secondary, and the turns ratio sets the secondary current approximately. A steady DC current can still create a magnetic field. After the switching transient, however, the flux stops changing and the secondary signal falls almost to zero. Removing the suspected cause and then restoring it is stronger evidence than changing several controls together.

TAKEAWAY 17W

A current transformer responds to changing magnetic flux; a steady primary current produces no sustained secondary signal.

8.3 The peak, and how far the cold moves it

LOAD & FORECASTING · A PERSON · BALLPARK

SCENE 36W

The base peak for a Friday in this week of the year is 6,400 megawatts. Lindgren's own figure for this winter is 120 megawatts of extra demand per degree below 0, and tomorrow is minus 4.

GUIDE 63W

Four numbers, and one of them is the reserve held, which belongs to the width of the forecast rather than its centre. Ask of each whether it is a base, a sensitivity, or a temperature. A sensitivity is megawatts per degree on this system, measured from this winter. What comes out is the centre of a forecast, and the reserve covers the rest.

ASK 4W

Estimate tomorrow's evening peak.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

demand rises as the temperature falls, roughly in step · electrical power and energy over time — taken as read

VERDICT 79W

A load curve is the same shape most days, because people do the same things at the same times. What moves it is weather, and the useful form of that is a sensitivity: how many megawatts one degree is worth on this system. Measured from this winter's own data rather than from a textbook. Base plus sensitivity times degrees gets the centre of the forecast. It does not get the width, which is what the reserve has to cover.

TAKEAWAY 17W

A forecast is a base shape plus a sensitivity, and the sensitivity is measured rather than assumed.

END OF THE DAY 25W

Once source voltage and fault type are fixed, network impedance controls how large fault current becomes; that is why protection has to act in milliseconds.

DAY 9 OF 15

The valley breaks away

PLAN CARD 111W

Friday, 18:20. The route into the valley tripped at the worst hour of the week. It is the same route the room pushed hard a week ago, using a heat record nobody had checked. The valley did not go dark. It broke away from the rest of the system, and it is now on its own with one gas machine and the wind ridge. Reyes has minutes to choose. She can help the valley survive on its own, or shut it down on purpose. Nadia Haddad says the one machine in there has very little spare power behind it. Today you decide whether the valley keeps its power or loses it.

BRIEFING 21W

The Calder corridor trips at the evening peak; the valley is still lit, but it is now electrically on its own.

WHAT YOU DECIDE 21W

Use the island frequency to decide whether Calder Valley can stay alive on its own, and shed load before it collapses.

9.1 Why the island's frequency is falling slowly

SYSTEM OPERATIONS · A ROOM · DEGENERACY

SCENE 36W

The island frequency is falling at 0.10 hertz per second. Its dispatchable machine is already at full output, while inverter-connected ridge generation is low. Haddad wants the power shortfall, but the wall gives only the slope.

GUIDE 65W

Two controls slide along a family of combinations that all reproduce the observed 0.10 hertz per second. Find out how wide that family is before naming a number. The wall gives only the slope, and a slope alone cannot separate a large shortfall on a heavy system from a small one on a light system. Then bring in the stored kinetic energy, which collapses it.

ASK 18W

What power shortfall is consistent with the 0.10 Hz/s fall once the island's stored kinetic energy is measured?

BEHIND THE BUTTON 136W

Why the slope alone is not enough. Frequency falls at a rate set by the power shortfall divided by the stored kinetic energy of everything spinning. One equation, two unknowns. Every pair that gives the same ratio produces exactly the same slope on the wall.

Why the ridge generation matters. Inverter-connected machines contribute little or no stored kinetic energy — they are not spinning masses locked to the frequency. So an island leaning on them has less inertia than its megawatts suggest, and the same shortfall drives the frequency down faster.

What measuring the inertia buys. It fixes one of the two unknowns, so the shortfall follows immediately. That is why system operators now measure inertia rather than assuming it: the assumption was safe while the fleet was synchronous and stopped being safe as it changed.

TAKES AS READ

the island is near 50 Hz · the initial frequency slope follows $df/dt = f_0\Delta P/(2E\_k)$

VERDICT 81W

The initial frequency slope comes from two quantities at once: the real-power mismatch and the stored kinetic energy resisting it. A slow fall can therefore mean a small shortfall, large inertia, or a matching combination of both. Inverter-connected wind can add real power without adding the same synchronous kinetic energy, so installed megawatts do not settle the question. First trace the combinations that reproduce 0.10 Hz/s. Then use an independent estimate of stored kinetic energy to select the matching power shortfall.

TAKEAWAY 21W

A frequency slope constrains the ratio of power imbalance to stored kinetic energy; it does not determine either one by itself.

9.2 What is stored, and for how long

METERING & STANDARDS · A PERSON · PROTOCOL

SCENE 30W

Farrow is asked to list what the island has to draw on in the first ten seconds. Two of the four things on her list cannot help at that timescale.

GUIDE 58W

Four stores of energy, and the right column is four timescales. Pair them by asking how soon each one answers, and for how long. Two of these are on the list and cannot help in ten seconds. The fields in transformers and lines hold real energy for a fraction of a cycle, which is nothing you can dispatch.

ASK 9W

Say where an island's short-term energy actually comes from

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

energy can be stored in a spinning mass, in a magnetic field or in a chemical cell · aC waveforms: frequency, period and phase — taken as read · charge, current, voltage and resistance — taken as read

VERDICT 81W

Each store answers over a different interval and confusing them is how a plan fails. Rotational energy in a spinning machine is there instantly, without anybody asking, and it is gone in seconds. A battery through an inverter responds in well under a second and can hold for hours. That makes it the most useful thing an island can have. Energy in the magnetic fields of transformers and lines is real, and it is a fraction of a cycle: nothing dispatchable.

TAKEAWAY 16W

> Every store has a timescale, and matching the store to the timescale is the whole trick.

9.3 Keeping it, or collapsing it on purpose

DISTRIBUTION · DISTRIBUTION DEPOT · A ROOM · CHOICE

SCENE 37W

Obi has 26,000 households inside the island and a frequency still falling. Holding it needs demand off. Collapsing it means all of them, and a longer way back. The event board is still showing the overnight state.

GUIDE 60W

Four options, and the island has one dispatchable machine already at maximum. So ask what lever each option actually uses. Two of them are bets on something outside the room. One costs all twenty-six thousand households and a restoration measured in hours, because a dead network is built back piece by piece. The frequency is still falling while you choose.

ASK 7W

What should be done with the island?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

restoring a collapsed island takes longer than holding a live one

VERDICT 78W

An island with its only dispatchable machine at maximum has one lever left, and that is demand. Shedding a fifth stops the fall and keeps four fifths of the valley supplied, which is a far better outcome than everybody off. Letting it collapse costs all 26,000 households and buys a restoration measured in hours. That is because a dead network has to be built back piece by piece. Waiting on wind is betting the island on a forecast.

TAKEAWAY 16W

Shedding some load to keep an island alive is usually cheaper than restoring it from nothing.

9.4 Loss when the load turns up anyway — Review

TRANSMISSION & PROTECTION · A ROOM · CALLBACK · BALLPARK

SCENE 40W

Thabo Dube, the substation technician, is standing at the second circuit with a clamp meter and a drawing. It has picked up flow it does not normally carry, and Piotr Novak, the protection engineer, wants the heat figure before dark.

GUIDE 65W

Five numbers, and two of them are about the circuit rather than about tonight. The current it normally carries belongs to a comparison you can make afterwards. The operating voltage belongs to a different relationship entirely. Ask of each number whether the heat depends on it. Then notice that the flow has doubled, and decide what that does to a quantity the current enters twice.

ASK 11W

Estimate the power now lost as heat in the second circuit.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

loss in a conductor rises with the square of the current · a thermal rating is a temperature reached over a stated time · ohm's law and resistive networks — taken as read · electrical power and energy over time — taken as read

VERDICT 85W

Resistive heating in one phase is the current squared times the resistance, and three phases carry three times that. This circuit is holding twice its usual flow, so it sheds four times its usual heat — about thirteen megawatts against a little over three. What matters is what that does to the margin. A conductor's limit is a temperature reached over time, and it is the heat that drives the temperature, so a doubled current spends the margin far faster than the current figure suggests.

TAKEAWAY 21W

A circuit standing in for another does not share out the heat in proportion to the flow it has taken on.

END OF THE DAY 16W

An island lives or dies on whether generation inside it can follow the load inside it.

DAY 10 OF 15

The lights go out

PLAN CARD 98W

Friday, 19:04. The load shed at 18:20 kept most of Calder Valley alive for forty-four minutes. Then the ridge wind fell faster than forecast, the remaining shortfall returned, and the island collapsed. Now the valley is dark. The gas station cannot start because its pumps, fans and controls all need electricity first. Haddad has one small hydro unit above the valley that can start with no outside power. Dube has a yard full of open switches and no safe sequence yet. Today you build a live system out of a dead one, one energised link at a time.

BRIEFING 24W

The load shed bought forty-four minutes. Then the wind fell, the island collapsed, and every large machine lost the power it needs to restart.

WHAT YOU DECIDE 17W

Rebuild Calder Valley from the one hydro unit that can start without power from the dead grid.

10.1 What the dark hours actually cost

GENERATION · GENERATION HALL · A ROOM · BALLPARK

SCENE 33W

The valley was drawing about 300 megawatts when it went. Haddad expects the restoration to take two and a half hours, and the report will want the figure in energy rather than power.

GUIDE 61W

Four numbers, and two of them belong to other questions: the household count and the frequency before the split. Ask of each whether it is a power, a duration, or neither. Power says how fast energy is delivered. Energy says how much arrived, and it is what a settlement runs on. Both get called demand, which is where this goes wrong.

ASK 8W

Estimate the energy not delivered during the restoration.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

power sustained over a period of time delivers energy · charge, current, voltage and resistance — taken as read

VERDICT 84W

Power says how fast energy is being delivered. Energy says how much arrived. The two are confused constantly, because both get called demand. Multiply the power by the time it was sustained and the result is energy. Megawatt-hours, if the power was in megawatts and the time in hours. That is the number every settlement, every compensation claim and every inquiry runs on, because it is the one that scales with duration. Two systems can lose the same megawatts and owe entirely different amounts.

TAKEAWAY 16W

Power is a rate and energy is the amount, and the settlement is always in energy.

10.2 The order physics allows

SYSTEM OPERATIONS · A ROOM · CHAIN

SCENE 32W

One hydro set that can start unaided, a dead network, and a room full of people who want the big station back first. Reyes needs the sequence written before anybody switches anything.

GUIDE 64W

Build the path in the order the energy actually travels, from the machine that can start unaided to the one that cannot. Then name the required link that limits it. The biggest station is not the answer, and neither is the largest number on the board: a path is limited by the link with the least to give, and the readings tell you which.

ASK 15W

How does startup power reach the large station, and which required link limits the path?

BEHIND THE BUTTON 139W

What black start means. Almost every generator needs power to start: excitation, auxiliaries, pumps, controls. A hydro set with a battery, a governor and a head of water behind it does not, which makes it the only place a dead network can begin. Everything else is downstream of it.

Why the order is not a preference. Energising a path means charging line capacitance and picking up auxiliaries in an order the equipment can survive. Take too much at once and the small starting machine collapses, and the network is dead again with less to try next.

Why the room wants the wrong thing. The big station is what everybody is waiting for and what the public will notice. That is a reason to write the sequence down before anybody switches anything, rather than a reason to put it first.

TAKES AS READ

the large station needs electrical auxiliaries before it can start · the hydro unit can start without an outside electrical supply

VERDICT 87W

A dead network cannot start from its biggest generator if that machine needs pumps, fans and controls before it can turn. The black-start hydro unit works because water can supply torque without outside electrical power. Its generator then energises a bus and a narrow startup feeder to the large station. The feeder only needs to carry the station auxiliaries, not the future 500 MW output. That modest link governs the restart. Once the auxiliaries are live, the large machine can start and the system can grow outward.

TAKEAWAY 24W

A black start is a dependency chain: the largest generator is useless until a smaller source can energise its auxiliaries through a complete path.

10.3 Why it cannot go faster

DISTRIBUTION · DISTRIBUTION DEPOT · A PERSON · CASEBOOK

SCENE 35W

Obi has crews asking why restoration is moving a few streets at a time. The substation is live, and the whole feeder could be closed in one movement. The switching sheet is open beside her.

GUIDE 57W

Four restoration steps, and four different limits. Pair them by asking what is small: the live system, the load on the line, or the agreement between two systems. One of these is the opposite of most people's instinct. A long circuit with almost nothing on it raises the voltage at its far end rather than lowering it.

ASK 10W

Match each restoration step to the reason it is limited

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a small live system is disturbed more by a given block of load than a large one

VERDICT 89W

The limits change as the system grows. Very early, the live island is tiny and a block of load is enormous relative to it. Blocks stay small: this is a frequency limit. Energising long circuits with almost no load on them raises voltage rather than lowering it. That is a reactive limit and the opposite of most people's instinct. Cold load picks up higher than normal, because every thermostat in the street is calling at once. And synchronising two live systems is a matching problem, not a capacity one.

TAKEAWAY 16W

Every restoration step is limited by something different, and hurrying the wrong one collapses it again.

END OF THE DAY 18W

A dead system is rebuilt from one machine that needs nothing, outward along a path chosen in advance.

Six hours, not forty minutes

PLAN CARD 100W

Sunday. Farrow has finished checking the heat records for the route into the valley. The number that bought the room forty minutes came from one sensor. That sensor was reading eleven degrees too low, and had been since a software update in the spring. So the route was never forty minutes from its limit. It had already been over its safe limit for six hours, and twice before that. Every call made this week used that number somewhere. Today you decide which of those calls still stand. Nobody reasoned badly. Everybody reasoned from a number, and the number was wrong.

BRIEFING 23W

Farrow's independent checks show the corridor sensor was reading 11 °C low; the room's supposed forty-minute margin was built on the wrong temperature.

WHAT YOU DECIDE 15W

Re-read the blackout with the corrected corridor temperature and determine which earlier decisions still stand.

11.1 The sensor that was confident and wrong

METERING & STANDARDS · A ROOM · CHOICE

SCENE 31W

Farrow has three sources for the corridor's temperature that week: the sensor, an infrared survey from the ground. The sag measured off a photograph. Two of them agree with each other.

GUIDE 55W

Four options, and three temperature sources have to be accounted for. Two agree with each other and neither of them shares the corridor sensor. Ask of each option how many of the three it explains. Small scatter shows an instrument repeats, not that it is right. And the firmware date marks where the disagreement starts.

ASK 8W

Which explanation best fits all three temperature sources?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

an instrument can be precise and still be offset from the truth

VERDICT 86W

Small scatter only shows that an instrument repeats. A fixed offset can produce readings that are precise and wrong at the same time. Here 2 methods that do not share the corridor sensor both place the conductor about 11 degrees hotter. The firmware date also marks when the disagreement begins. Sag depends on tension as well as temperature, so it is not proof alone. Its value is that it corroborates an infrared method through different physics, which is exactly what a sensor offset cannot explain away.

TAKEAWAY 17W

> Precision is repeatability; accuracy is being right. A drifted sensor has the first and not the second.

11.2 What a rating is a statement about

TRANSMISSION & PROTECTION · A ROOM · SWEEP

SCENE 50W

The corridor's continuous operating limit corresponds to a conductor temperature of 75 °C under the stated weather assumptions. Above that, sag increases immediately, while sufficiently hot or prolonged operation can also accelerate creep and annealing. Novak wants the current that reaches the 75 °C operating limit under the corrected conditions.

GUIDE 67W

Drag the corridor current. The readout is the conductor's own temperature, not its rating, and only the currents you stop at get plotted. You want the current at which that temperature reaches the 75-degree limit named above. Heating rises faster than the current causing it, so the curve steepens as you go and the last hundred amps cost more than the first. Mark that current and commit.

ASK 20W

75 degrees is the limit the conductor was built to. What current does that allow it to carry all day?

BEHIND THE BUTTON 196W

What sets the temperature. Current heats a conductor as the square of itself, and the conductor loses that heat to the air by convection and to the sky by radiation. The temperature settles where the two balance, which is why the curve is not a straight line: doubling the current asks the cooling to remove four times as much heat.

Why a rating is a temperature. The ampere figure printed on a corridor is the current that reaches the temperature limit under an assumed ambient, an assumed wind and an assumed sun. Change any of those and the same amps arrive at a different temperature. That is why a corrected weather figure can move a rating without anybody touching the line, and why a rating quoted without its assumptions is not a limit at all.

What happens above the limit. Aluminium held above its continuous temperature begins to anneal, losing strength it does not get back, and the hotter span stretches and sags. The clearance over the road below is a legal minimum, so the temperature limit is doing two jobs at once: protecting the conductor, and keeping the line where the survey says it is.

TAKES AS READ

a conductor's rating is a limit on temperature rather than on current · resistive heating rises faster than the current that causes it

VERDICT 75W

A rating is not a fuse setting. It is the current that produces an agreed conductor temperature under assumed ambient conditions. The heating rises roughly with current squared, while cooling depends on wind, air temperature, radiation and the conductor itself. The corrected curve reaches 75 °C near 1,150 A and puts 1,280 A near 89 °C. That proves the operating limit was exceeded; it does not, by itself, measure how much permanent material damage occurred.

TAKEAWAY 25W

A thermal rating is a temperature translated into current under stated weather assumptions, so bad temperature data erase the operating margin you thought you had.

11.3 What survives a bad number

SYSTEM OPERATIONS · A PERSON · CASEBOOK

SCENE 36W

Reyes has the week's decisions on the board and one wrong input underneath several of them. She wants each one marked as still good, now unsupported, or wrong. The yard radio is still live beside them.

GUIDE 59W

Four decisions from the week, and one wrong input underneath some of them. Trace each decision back to what it was argued from. Ask whether temperature is in that chain at all. A loss calculation that used current and resistance never touched the sensor. Neither did a frequency. One wrong number voids what leaned on it and nothing more.

ASK 8W

Decide which of the week's conclusions still stand

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a conclusion can depend on one input or on several

VERDICT 85W

One wrong number does not void a week. It voids whatever leaned on it, and the job is tracing which. The choice to hold the corridor rested entirely on the duration, so it falls. The loss calculation used current and resistance and never touched temperature, so it stands. The restoration order was argued from consequence and household counts, which the sensor never entered. And the decision to shed a fifth of the island was taken on a frequency the sensor has nothing to do with.

TAKEAWAY 20W

A bad input invalidates what rested on it and leaves the rest standing, and telling them apart is the work.

11.4 What can come off, and what cannot

LOAD & FORECASTING · A ROOM · BELT

SCENE 34W

A shed order needs a list in ten minutes. Some feeders can be dropped and restored an hour later with nobody harmed. Others carry something that cannot simply be switched off and switched back.

GUIDE 57W

Two bins. A feeder can be shed if the load behind it tolerates an hour without power and comes back the way it went off. It cannot be shed if something behind it needs power to stay safe, or cannot be restarted at will. Sort on what is behind the feeder, not on how important it sounds.

ASK 15W

Send each feeder to the bin that says whether the shed order can take it.

BEHIND THE BUTTON 97W

Why sheddable is not the same as unimportant. A large industrial customer with its own standby plant is sheddable; a small clinic on one supply is not. Size, revenue and prominence have nothing to do with it, which is why the list is written in advance and not argued out at the time.

What makes a load unsheddable. Something behind it that fails unsafely without power, or that cannot be brought back on demand — a furnace that sets solid, a pumped drain that floods, a process that has to be run to a finish once started.

TAKES AS READ

some loads can be interrupted and restored, and some cannot

VERDICT 141W

A shed order is written to the state of what is behind each feeder. On one side sit the loads that tolerate an hour off and come back exactly as they went — cold stores with thermal mass, offices, street lighting, a works with its own standby set. On the other side are loads where an hour without power is not an hour of inconvenience: a hospital on one supply, a smelter whose bath sets solid, a pumped drain under a town, a process that has to run to a finish once it has started. None of that maps onto size or importance, which is exactly why the list is prepared cold and not decided at four in the morning by whoever is loudest on the phone. The one thing a dispatcher cannot do is work it out during the event.

TAKEAWAY 18W

What can be shed is decided by what is behind the feeder, not by how important it sounds.

END OF THE DAY 23W

A conclusion is only as good as the measurement under it, and a drifted instrument leaves the reasoning intact and the answer wrong.

The quiet shift

PLAN CARD 105W

Monday, and nothing is going wrong. The forecast is holding to within thirty megawatts. Every line is well inside its limit. The wind ridge is steady, and there is nothing on the board to decide. Aaron Whitlock, the assistant operator, has almost nothing to write in the log. Reyes walks the room on evenings like this and asks people what they are assuming, because a quiet shift is the only time there is. Today you sort tonight's comforts into two piles: the ones somebody measured, and the ones everybody has got used to. Nothing will go wrong tonight. Nothing went wrong two Mondays ago either.

BRIEFING 21W

Nothing is wrong tonight, which makes this the only cheap time to find out which comforting numbers are only inherited assumptions.

WHAT YOU DECIDE 20W

Use a calm night to identify the assumptions that could change the next emergency before an emergency forces the test.

12.1 The failure that has not happened

SYSTEM OPERATIONS · A ROOM · CHOICE

SCENE 38W

The screen says every single contingency is survivable tonight. Whitlock asks what the check does not cover, and Reyes says that is the right question on a quiet shift. The event board is still showing the overnight state.

GUIDE 58W

All four options describe something a clean contingency check might mean. They differ in scope: one failure or two, survivable or likely. Ask of each whether the check's own list could support it. The check is exactly as broad as that list. Last Friday was a corridor already above a limit nobody knew about, and then a trip.

ASK 9W

What does a clean contingency check tell you tonight?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a system is required to survive any one credible failure

VERDICT 120W

The check is exactly as broad as its list. It takes each credible single failure in turn, works out the flows afterwards, and confirms nothing ends up over a limit. That is a genuinely strong statement and a narrow one. It does not cover two failures at once, it does not cover a failure outside the list. It says nothing at all about what is likely — only about what is survivable. Last Friday was two things: a corridor already above a limit nobody knew about, and then the trip. When a second thing lands on neighbours checked only one at a time, each trips correctly in turn, and the sequence of events is what has to be reconstructed afterwards.

TAKEAWAY 15W

> An N-1 check is a statement about single failures, and it says nothing about 2.

12.2 The trend with nothing wrong with it

LOAD & FORECASTING · A PERSON · PROTOCOL

SCENE 34W

Lindgren's forecast has tracked within 30 megawatts all evening. He wants to know what that does and does not license the room to assume about tomorrow. The room is waiting for a written instruction.

GUIDE 64W

Four observations about a forecast, and the right column says what each one licenses. Pair them by asking where in the distribution that day sits. A mild Monday is the middle, and the middle is where the model was best calibrated. Only one of these four is evidence about the method rather than about a night, and the room is deciding reserve on it.

ASK 6W

Read a forecast that is behaving

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a forecast can be right by luck as easily as by skill on any one day · charge, current, voltage and resistance — taken as read

VERDICT 77W

A forecast is a distribution. Landing inside 30 megawatts on a mild Monday is what the middle of that distribution looks like and it tells you almost nothing. That is because mild Mondays are the days the model was best calibrated for. The evidence that matters is how it behaves on the days it was least calibrated for, which are rare by construction. So a run of good nights is not a reason to hold less reserve.

TAKEAWAY 18W

> One good night is not evidence about the method, because the width was never a claim about tonight.

12.3 Which of these is a measurement

GENERATION · GENERATION HALL · A ROOM · VALUE

SCENE 30W

Haddad lists four reasons the room feels comfortable tonight. Three are current measurements or long-run evidence. One is an inherited stored-energy figure written before the inverter-connected fleet became much larger.

GUIDE 52W

Four reasons the room feels comfortable, and four check credits. Open each one and ask what it actually is: a current measurement, long-run evidence, or a number somebody inherited. Buy the evidence that could change the under-frequency timing. A figure written before the fleet changed is a claim about a different system.

ASK 13W

Which evidence should be bought before the room trusts its present under-frequency timing?

BEHIND THE BUTTON 144W

Why the stored-energy figure is the suspect. Under-frequency protection is timed against how fast the frequency can fall, and that depends on the inertia of everything spinning. As inverter-connected generation grew, the inertia fell — but a number written down before that shift keeps being quoted because it has been in the file for years.

What separates a measurement from a comfort. Three of the four can be pointed at something measured recently on this system. The fourth is an inheritance, and the room's confidence rests on it without anybody having decided to trust it.

Why timing is where it bites. Under-frequency load shedding has to act before the frequency reaches the point where machines trip. If the real inertia is lower than assumed, the frequency gets there sooner and the settings are late — which is a blackout rather than a load-shedding event.

TAKES AS READ

stored kinetic energy affects the initial rate of frequency change · inverter-connected generation may add power without adding synchronous inertia

VERDICT 71W

Three comforting numbers are current measurements or long-run evidence. The inherited inertia figure is different: the fleet has changed, and inverter-connected generation can supply real power without the same directly coupled synchronous kinetic energy. Recomputing stored kinetic energy from current machine data therefore changes the expected rate of frequency fall. If the old 25,000 MW·s estimate has fallen to 18,000 MW·s, the same 300 MW trip moves frequency about 40% faster.

TAKEAWAY 18W

An old assumption deserves priority when it is load-bearing, likely to have changed, and cheap enough to check.

12.4 What the desk is chasing now

DISTRIBUTION · DISTRIBUTION DEPOT · A ROOM · SPOT

SCENE 42W

Faults arrive on the board all evening. The desk works to one standing priority, and the shift manager changes it as the picture changes — after the storm crosses, after the hospital calls, after the count of customers off passes ten thousand.

GUIDE 55W

Take the faults the standing priority currently wants and leave the rest. The priority at the top of the board is replaced during the evening and nobody announces it. What is scored is the faults either side of a change, because they are the only ones that show whether the desk is reading the board.

ASK 12W

Dispatch to the priority on the board, and keep watching the board.

BEHIND THE BUTTON 97W

Why the priority changes at all. Early in an event the work is restoring the most customers per crew-hour. Once a hospital is on generator the work is that hospital. After midnight it is whatever cannot be left overnight in the weather. Each of those is a different order of the same list.

Why the changeover is where it goes wrong. A desk that keeps dispatching to the old priority for ten more minutes sends its last free crew to the job that mattered an hour ago, and nothing on the board says that is what happened.

TAKES AS READ

a control desk works to a stated priority that somebody can change

VERDICT 121W

Three priorities run across the evening. Each one wants a different part of the board. Most customers restored first. Then anything with a vulnerable customer behind it. Then anything that cannot be left out overnight in the weather. A fault can answer two of those at once, and that is what makes the switch cost real. What the panel scores is the window either side of each change. Everything else on the board is wanted by neither priority, so it is left alone correctly by somebody who has read nothing. The desk's own version is the ten minutes after a hospital calls. The priority has already changed, and the crews rolling are rolling to the order in force when they left.

TAKEAWAY 15W

The cost of a withdrawn instruction is paid by whoever is still working to it.

END OF THE DAY 15W

A quiet system is a set of assumptions nobody is currently being forced to check.

Cheapest is not enough

BEFORE THE DAY — CANVASS

Whose lights are actually back

The system says eleven feeders are restored. The complaint line says otherwise, and the difference matters because a feeder that reads energised at the station and dead at the street is a fault somewhere in between. Ask along the affected streets until enough answers have come back to tell those two possibilities apart.

PLAN CARD 102W

Thursday. Tomorrow's power has to be bought tonight. Lindgren has a peak that moves with the temperature and a range he will not narrow. Mina Sarraf, the wind fleet controller, has a ridge that costs almost nothing to run and cannot promise anything. She is tired of being asked to promise. Nadia Haddad has expensive machines that do exactly what they are told. Reyes has to sign an order that puts them in a sequence. Today you decide what cheapest really means when the cheapest power is the least certain. The ridge may also make more power than the lines can carry.

BRIEFING 19W

Tomorrow’s power must be scheduled tonight; the cheapest fleet is variable, and the Calder corridor is still the bottleneck.

WHAT YOU DECIDE 18W

Choose tomorrow’s generation and reserve when the cheapest power is uncertain and the corridor cannot carry everything available.

13.1 The ground is not at one voltage

DISTRIBUTION · DISTRIBUTION DEPOT · A ROOM · CHOICE

SCENE 38W

The cable is isolated at both ends and a tester reads dead at the work point. Kowalczyk, the safety lead, stops the crew because one protection step is still missing. The substation remains connected to live equipment nearby.

GUIDE 59W

All four options are things a crew could do next, and three of them sound careful. Ask of each what it would protect against: a source you know about, or one that arrives later. Isolation removes the known sources. A dead test confirms one moment. Neither holds the conductor down if switching changes or a voltage is induced nearby.

ASK 10W

What must happen after isolation and a successful dead test?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a conductor can be re-energised from more than one direction · charge, current, voltage and resistance — taken as read

VERDICT 81W

Isolation and proving dead answer different questions. Opening both ends removes the known sources, while the dead test confirms the condition at that moment. Neither guarantees the conductor will stay safe if switching changes or induced voltage appears. Portable earths provide a low-impedance path for unexpected current and keep the work conductor close to local ground potential. That is why earthing follows the dead test rather than replacing it. The sequence turns a measured safe state into a maintained safe state.

TAKEAWAY 13W

Isolation is not proof of safety, and earthing is what makes it provable.

13.2 Cheapest first, and the exceptions that are not exceptions

SYSTEM OPERATIONS · A PERSON · CASEBOOK

SCENE 30W

Four plants and an order to sign. Reyes wants each one matched to the actual reason it sits where it does in the stack, rather than to its price alone.

GUIDE 64W

Four plants and four reasons to sit where they do. Only one of the four reasons is price. Pair them by asking what bends the merit order: reliability, the ability to move quickly, or where the plant is. A machine at full output cannot sell reserve. A cheap machine in the wrong place can be worse than an expensive one in the right place.

ASK 11W

Match each plant to why it is dispatched where it is

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

generation is scheduled in order of what it costs to run

VERDICT 89W

Dispatch starts from cost, because running the cheapest first is what makes electricity affordable. Then three things bend it. Plant with almost no running cost goes first and cannot be relied on. It displaces fuel without displacing the need for something reliable behind it. Plant that can change output quickly is worth holding part-loaded, because reserve is a service and a machine at full output cannot provide it. And plant in the right place can be needed regardless of price, when the alternative is a corridor over its limit.

TAKEAWAY 15W

Merit order is cost order, corrected for what the system also needs from a machine.

13.3 When the cheapest power cannot get out

GENERATION · GENERATION HALL · A ROOM · SCIENCETANK

SCENE 40W

Sarraf can produce 340 MW from the ridge tonight, but the Calder export corridor can carry only about 190 MW under the current operating limit. There is no other export path available. A printed network diagram lies on the desk.

GUIDE 100W

The ridge can produce 340 MW tonight and the corridor is limited to about 190 MW under the current operating case. Nights like this occur perhaps forty times a winter. Storage at the ridge would be 50 MW for four hours at 85% round-trip efficiency, moving surplus in time rather than in space. A third corridor circuit has been on the reinforcement list for two years without a construction window. Curtailment payments recur every constrained night. Operating hotter than the continuous case increases sag and can add cumulative thermal damage, so exceeding the limit is not a free fifth option.

ASK 12W

Decide what to do with more wind than the corridor can carry

BEHIND THE BUTTON 193W

Why the whole spread is graded. Funding is not a vote for one idea. A portfolio says what you think is likely, what is worth hedging against, and what is not worth pursuing at all — and the last two are where most of the information is. Backing the right proposal while quietly funding a bad one is a worse answer than it looks. That is why the small numbers count as much as the big one.

What the three numbers are for. Thirty-five is what makes a lead a lead: below it you have hedged rather than chosen. Fifteen is the most that can sit on unsupported work before it stops being a rounding error. Past that it is a second opinion nobody argued for. Twenty is the floor under a line of work you have already called strong, because funding it too thin to finish spends the money and buys nothing.

Why there is a floor on the total. Points held back are not caution; they are a decision not to decide, taken with somebody else's money and somebody else's deadline. The floor is what forces the panel to say something.

TAKES AS READ

power has to have a path to where it is used

VERDICT 69W

Generation that cannot reach customers is not usable output for the system. The corridor limit is therefore an operating constraint, not a preference. Some ridge output must be curtailed unless storage can move it to a later hour or transmission can be reinforced. Storage moves surplus in time; another circuit increases the path available in space. Running hotter simply spends thermal and clearance margin and can add cumulative damage.

TAKEAWAY 12W

> Curtailment is what a network limit looks like from the generator's end.

13.4 Fifty hertz, through the evening rise

SYSTEM OPERATIONS · A ROOM · HOLD

SCENE 29W

The evening rise is starting and the frequency is drifting under it. Governors are answering, and the only thing left on the desk is the regulating set's raise-lower control.

GUIDE 62W

Hold the frequency inside the band on the wall display. The band narrows as the evening goes on, because the tolerance a system can live with at seven o'clock is not the one it can live with at peak. The raise-lower control is what you have, and every load that comes on is a mismatch that keeps pushing until generation answers it.

ASK 10W

Hold the system at fifty hertz through the evening rise.

BEHIND THE BUTTON 154W

Why frequency drifts at all. Frequency is the running balance between what is being generated and what is being drawn. A load that comes on is not a step in the frequency — it is a step in the *rate*, and the machines slow for as long as the mismatch lasts.

What the governors have already done. Droop has answered part of each step automatically and left an offset behind: that is what droop is for, and it is why the frequency settles low rather than returning. Bringing it back to fifty is the operator's job, and it is what this control is.

Why the band narrows. Early evening the system has inertia and reserve to spare. At peak, with less spinning and more load, the same excursion is closer to the settings that shed feeders — so the same skill has to be exercised more tightly on exactly the hour it is hardest.

TAKES AS READ

frequency falls when demand exceeds generation

VERDICT 164W

Every event here is a step in the rate rather than a step in the reading. A block of load coming on does not drop the frequency by twenty millihertz and stop; it makes the system lose speed and it goes on losing while the mismatch stands. That is why the raise-lower control has to be moved to a new position and left there rather than nudged and released, and it is the whole difference between operating a system and chasing a needle. Droop has already answered part of each step and left an offset — a system that settles at 49.94 has done exactly what it was designed to do — so returning to fifty is a separate action taken by somebody. And the band narrows for a reason that is not arbitrary: at peak there is less inertia spinning and less reserve behind it, so the same excursion sits closer to the frequency at which feeders start coming off on their own.

TAKEAWAY 13W

> A mismatch you do not answer is not a mismatch that goes away.

END OF THE DAY 18W

Dispatch orders plant by cost, and the value of certainty is what stops that being the whole story.

DAY 14 OF 15

The last morning

PLAN CARD 91W

Monday morning. The reinforcement decision comes back for the last time before the construction window closes at five. The next window is eighteen months away. Farrow's corrected record shows about ninety hours above the continuous operating limit over three years. That establishes repeated thermal exposure, not an exact remaining tensile strength. Alvarez has a storage quote that expires today, and the board will not fund every option. Today you decide what can be justified now, what still needs verification, and whether the system spends another eighteen months managing the Calder bottleneck.

BRIEFING 23W

The construction window closes at 17:00; the corrected record proves repeated high-temperature operation, but the exact remaining conductor strength is still being verified.

WHAT YOU DECIDE 19W

Commit the corridor to a course before the construction window closes, using the corrected evidence rather than assumed damage.

14.1 Hours over limit, added up

METERING & STANDARDS · A ROOM · CHOICE

SCENE 37W

Farrow's corrected record has the corridor above its continuous operating temperature for about ninety hours over three years. She wants one sentence for the board: what does that record justify saying before a conductor sample is tested?

GUIDE 49W

Separate exposure from damage. Higher temperature increases sag immediately, and sufficiently hot or prolonged operation can produce irreversible creep or annealing. But the amount depends on conductor construction, temperature and time. The record proves repeated operation outside the continuous case; it does not directly measure the remaining tensile strength.

ASK 8W

What does the ninety-hour record justify saying now?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

thermal effects depend on both temperature and exposure time · an operating record can establish exposure without directly measuring remaining tensile strength

VERDICT 74W

The corrected history matters because thermal effects do not reset simply because the conductor cools between events. Higher temperatures increase sag immediately, and long or hot exposure can contribute to permanent creep or annealing. But the record is an exposure history, not a tensile test. The defensible conclusion is that the conductor’s remaining margin must be evaluated; claiming a specific permanent strength loss before modeling or testing would turn an inference into a measurement.

TAKEAWAY 19W

Cumulative high-temperature exposure is a design concern, but the operating record alone does not measure the conductor's remaining strength.

14.2 What the window is worth

TRANSMISSION & PROTECTION · A PERSON · SCIENCETANK

SCENE 44W

Two options, one budget, one afternoon. Novak points out that only one option removes the corridor bottleneck; Haddad points out that only one construction window disappears at five. The corrected thermal record is on the table, and the exact conductor-strength test is still pending.

GUIDE 76W

The construction window closes at five and the next one is eighteen months away. The corrected record already proves repeated operation above the continuous thermal case. The storage quote also expires today, but storage does not add transmission capacity. Curtailment on nights like last Friday occurs about forty nights a winter. Farrow's strength review will refine the asset condition, but it will not create another corridor path. The board will not approve both storage and reinforcement.

ASK 7W

Commit the money before the window closes

BEHIND THE BUTTON 193W

Why the whole spread is graded. Funding is not a vote for one idea. A portfolio says what you think is likely, what is worth hedging against, and what is not worth pursuing at all — and the last two are where most of the information is. Backing the right proposal while quietly funding a bad one is a worse answer than it looks. That is why the small numbers count as much as the big one.

What the three numbers are for. Thirty-five is what makes a lead a lead: below it you have hedged rather than chosen. Fifteen is the most that can sit on unsupported work before it stops being a rounding error. Past that it is a second opinion nobody argued for. Twenty is the floor under a line of work you have already called strong, because funding it too thin to finish spends the money and buys nothing.

Why there is a floor on the total. Points held back are not caution; they are a decision not to decide, taken with somebody else's money and somebody else's deadline. The floor is what forces the panel to say something.

TAKES AS READ

an option with an expiry date is worth more than the same option without one

VERDICT 70W

This is different from last week because one option now expires. A third circuit is the only proposal that removes the transmission bottleneck rather than working around it. Waiting eighteen months means more curtailment and more operation close to a thermal limit whose historical margin was overstated. The pending strength test should change maintenance assumptions if needed, but it does not erase the measured congestion or create a new path.

TAKEAWAY 14W

> When one path closes today, its value includes everything the eighteen-month delay would cost.

14.3 The 18 months you just bought

SYSTEM OPERATIONS · A ROOM · PROTOCOL

SCENE 31W

Reyes wants the consequences written down before the decision is signed, so that the next shift inherits a plan rather than a mood. The room is waiting for a written instruction.

GUIDE 69W

Four things on the left, and what the decision does to each on the right. Pair them by asking what changes today and what changes only when the circuit exists. A build is a date, not relief. Between now and then the corridor is as weak as it was this morning, and now known to be weaker. One of these is closed so a shift can act at 16:40.

ASK 8W

Say what the decision commits the room to

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a decision to build something is also a decision about the years before it exists

VERDICT 70W

A build is not relief. It is a date. Between today and that date the corridor is exactly as weak as it was this morning, and now known to be weaker. The room inherits three years of damage and no new capacity. So the emergency rating is recalculated on the corrected sensor. The curtailment nights are budgeted rather than absorbed. The contingency case is run at the corridor's real strength.

TAKEAWAY 22W

Committing to a fix commits you to running the system without it until it arrives, and that has to be planned too.

END OF THE DAY 20W

When a reversible option expires, the decision must use what is measured now while keeping unresolved asset-condition claims explicitly unresolved.

DAY 15 OF 15

Sign what you can prove

PLAN CARD 84W

Friday. The winter case is on Reyes's desk for signature. The sensor offset is well supported. The exact conductor strength is still being tested. One proposed relay cause is plausible but unproved. Lindgren will not let his winter peak be written as a single certain number. The last page asks which emergency practice becomes a permanent operating rule. Today you decide what the report is allowed to call a finding, where uncertainty must stay visible, and what rule the next night shift will inherit.

BRIEFING 23W

The winter case is due today; the final signature has to distinguish what is measured, what is inferred and what is still unknown.

WHAT YOU DECIDE 22W

Separate measured facts from inferences, verify the claims that still matter, and sign the one operating rule that should survive the emergency.

15.1 Measured, inferred, assumed

METERING & STANDARDS · A ROOM · ATTEST

SCENE 32W

June Farrow, the metering engineer, has four statements ready for the final report and only two verification slots left before sign-off. Some claims are already backed. Others are still inference or suspicion.

GUIDE 57W

Four statements are ready for the report and you have two verification slots. Open each claim and read what already backs it: some are measured, some are inferred from something measured, and some are suspicion written confidently. Hold what the backing does not support, and spend the checks where a wrong claim would do the most damage.

ASK 11W

Which claims deserve the limited verification budget before they become findings?

BEHIND THE BUTTON 149W

The three categories, and why they read alike. A measured claim points at a record. An inferred one is a chain of reasoning from a record, which is only as good as its weakest step. A suspicion has neither, and in a report all three appear as declarative sentences in the same typeface.

Why consequence rather than doubt decides the spend. The least-supported claim is not automatically the one to check. A weakly backed statement nobody will act on costs little; a well-worded inference that will drive an equipment replacement costs a great deal if it is wrong.

What holding a claim means. Not deleting it — marking it as not yet supported, so the report says what it can stand behind and what it cannot. A finding is a claim the author is willing to be wrong about in public, which is a higher bar than believing it.

TAKES AS READ

a report claim can be measured, inferred or still assumed · verification time is limited before the report closes

VERDICT 81W

The report contains claims at different evidence levels. The sensor offset is critical but already backed by two independent methods and a firmware date. Permanent conductor-strength loss is plausible metallurgy, yet no sample has tested it. The storage result is measured but not central to the cause finding. The proposed relay cause is both critical and still unproved. With only two verification slots, checking everything is impossible. Spend them on the critical claims whose present evidence cannot support a final finding.

TAKEAWAY 23W

A signed claim is still only as strong as the evidence behind it; verification should go first to critical claims that remain unbacked.

15.2 The number that has to be a range

LOAD & FORECASTING · A PERSON · CHOICE

SCENE 30W

Lindgren's winter peak goes into the report. He wants the width in the sentence, and Reyes wants to know what the width is for. The control-room phone is already ringing.

GUIDE 61W

Four reasons for a range, and they differ in what the width is for. Ask of each whether it is about blame, measurement, or planning. Reserve is not held against the expected peak. It is held against the amount by which the peak might exceed it, and that is the width. A single figure deletes the only quantity a planner needs.

ASK 16W

Why does the winter peak go in as a range rather than as a single figure?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a forecast is a distribution rather than a single value · charge, current, voltage and resistance — taken as read

VERDICT 81W

Reserve is not held against the expected peak. It is held against the amount by which the peak might exceed it. That is a property of the width and not of the centre. A report that states a single megawatt figure has deleted the only quantity a planner needs. The planner then invents one — usually badly, and usually optimistically. The width is also the honest part: it says which conditions the model was calibrated for and which it was not.

TAKEAWAY 16W

A prediction is only usable by a planner if it says how wrong it might be.

15.3 The rule you sign

SYSTEM OPERATIONS · A ROOM · CHOICE

SCENE 39W

Reyes turns the last page toward you. Four emergency practices are listed, but only one is cheap enough and powerful enough to become a standing rule on ordinary nights. Your choice is the final instruction in the winter case.

GUIDE 55W

All four changes are defensible and all four cost something every day. Ask of each whether it would have caught this campaign's failure, and whether the room will still be doing it in March. The sensor's offset survived months because nothing was compared against it. More frequent reports of a wrong number are not information.

ASK 9W

Which operating rule do you sign into permanent practice?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a practice adopted under pressure has a cost that only shows up later

VERDICT 75W

The sensor offset survived for months because no independent method was routinely compared with the number carrying the operating decision. A cross-check would have exposed that failure before the corridor crisis. Permanently elevated reserve is expensive and should instead be sized to forecast uncertainty. Faster reporting would only repeat the same bad sensor more often. A second signature on urgent load shedding adds delay to a decision that sometimes has to be made in seconds.

TAKEAWAY 23W

> Keep the practice that would have exposed the hidden failure before the crisis, without making every ordinary shift pay for an emergency-level response.

END OF THE DAY 20W

A signed finding is worth only the evidence under it; uncertainty belongs in the report, not hidden behind confident wording.

THE CLOSING CARD

At 19:04 Calder Valley went dark. The small hydro set above the valley started alone, energised the startup feeder, and gave the large station the pumps, fans and controls it needed to turn. Two and a half hours later the streets, pumps and clinic were taking power again. The black-start order you wrote worked.

Then Farrow's independent checks explained why the room had been so close to losing the valley. The corridor temperature sensor had been reading about eleven degrees low. The corrected history shows roughly ninety hours above the intended continuous-temperature case. That exposure is now being used to test the conductor's remaining margin instead of being turned into a strength claim the measurements cannot yet prove. The exact relay cause remains open.

You sign the Calder Winter Survival Plan with reserve sized to uncertainty, the third corridor circuit committed to the construction window, and one permanent operating rule: any instrument that carries an operating limit or emergency decision gets an independent check. Four million people finish the campaign with power, and the next night shift inherits a plan that says clearly what is measured, what is inferred, and what is still unknown.

Card text only — no options, boards or answers. 10,957 words of card prose across 52 stops. Generated from `themes/blackout` by `tools/make-cardbook.mjs`.